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19 April 1993

MILITARY SPECIFICATION

CARTRIDGE, CALIBER .50, M858 BALL AND M860 TRACER SHORT RANGE TRAINING AMMUNITION

This specification is approved for use by the U.S. Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, and preparation for delivery for Cartridge, Caliber .50, Ball and Tracer Short Range Training Ammunition (SRTA) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

O-I-503 - Insect Repellant
O-T-620 - Trichlorethane -1, 1, 1, Technical Inhibited (Methyl Chloroform)
P-D-680 - Dry Cleaning Solvent

| Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: AMSTA-AR-EDE, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. |

AMSTA FSC 1305

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FEDERAL (cont.)

VV-L-800 - Lubricating Oil, General Purpose Preservative, Water-Displacing, Low Temperature)

MILITARY

MIL-C-372 - Cleaning Compound Solvent for Bore of Small Arms and Automatic Aircraft Weapons
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-A-2550 - Ammunition, General Specification for
MIL-L-14107 - Lubricating Oil, Weapons Low Temperature
MIL-C-14858 - Cartridge, Caliber .50: Reference
MIL-L-46000 - Lubricant, Semi-Fluid (Automatic Weapons)
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-L-63460 - Lubricant, Cleaner and Preservative for Weapons and Weapon Systems

STANDARDS

MILITARY

DOD-STD-100 - Engineering Drawing Practices
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-636 - Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644 - Visual Inspection Standards and Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-650 - Explosives: Sampling, Inspection, and Testing
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
DOD-STD-973 - Configuration Management
MIL-STD-1168 - Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKING DRAWINGS

6006874	- "E" Cardboard Silhouette Target
6109086	- "F" Cardboard Silhouette Target
7643674	- Classification of Cartridge Case Defects Small Arms Ammunition
8421679	- M31A1, Trainfire Target Holding Mechanism
12576456	- Packing and Marking for Box, Wirebound for Cartridges, Caliber .50
12576457	- Packing and Marking for Box, Ammunition, M2A1 for Cartridge, Caliber .50
12002898	- "E" Polyethylene Silhouette Target
12002899	- "F" Polyethylene Silhouette Target
12598606	- Cartridge, Caliber .50, Ball, Short Range Training
12598607	- Cartridge, Caliber .50, Tracer, Short Range Training
* 19-48-4116/14	- Unitization Procedures for Boxed Ammunition and Components on Four Way Entry Pallets

INSPECTION EQUIPMENT DRAWINGS

LI12598606	- Index of Inspection Equipment Lists for: Cartridge, Caliber .50, Ball, Short Range Training
LI12598607	- Index of Inspection Equipment Lists for: Cartridge, Caliber .50, Tracer, Short Range Training

PUBLICATIONS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)

SCATP-5.56	- Ammunition Ballistic Acceptance Test Methods, Test Procedures for 5.56mm Cartridges
SCATP-7.62	- Ammunition Ballistic Acceptance Test Methods, Test Procedures for 7.62mm Cartridges
TB-700-2	- Department of Defense Explosives Hazard Classification Procedures
TECP 700-700 VOL. III	- Manual of Test Methods for Small Arms Ammunition
JSSPM	- Joint Services Safety and Performance Manual for Qualification of Explosives for Military Use

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TEST OPERATIONS PROCEDURES (TOPs)

U.S. ARMY TEST AND EVALUATION COMMAND (TECOM)

TOP 1-2-601 - Laboratory Vibration Schedule
TOP 3-2-045 - Automatic Weapons, Machine Guns, Hand and Shoulder Weapons
TOP 4-2-601 - Drop Tower Tests for Munitions
TOP 4-2-602 - Rough Handling

(Copies of other Government documents, drawings and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.
Orders should specify, "49 CFR 100-199 (LATEST REVISION")."

2.2 Nongovernment publications. The following document(s) forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

American National Standards Institute (ANSI)

ANSI Y14.5M-1994 - Dimensioning and Tolerancing

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained. (See contrast provisions for additional precedence criteria).

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3. REQUIREMENTS

3.1 General. The cartridges shall comply with the requirements shown on drawings 12598606 and 12598607, reference specifications and the requirements stated in this section. All of the cartridge components shall be compatible with each other. The primer mix shall be noncorrosive and the propellant shall be smokeless. The cartridge shall be compatible with the Caliber.50 M9 link and the M2 Browning Machine Gun (BMG) with the M3 Recoil Amplifier (RA) weapon system.

3.2 Qualification. A sample shall be submitted for qualification in accordance with (IAW) the technical provisions specified herein (see section 4). The purpose of qualification is to verity that a particular contractor's design shall be considered qualified indefinitely. The contractor shall submit level 3 detailed item drawings IAW DOD-STD-100, MIL-A-2550 and ANSI Y14.5-82 to the government for configuration control. All changes shall be submitted IAW DOD-STD-973 for review and approval by the government. The government reserves the right to subject the revised design to any or all the qualification evaluations specified in 4.3 at the expense of the contractor.

3.3 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection IAW the technical provisions specified herein (see 4.4).

3.4 Performance characteristics.

3.4.1 Primer sensitivity. The value of the average height (h) minus 2 standard deviations (s) shall be equal to or more than 63mm (2.5"). For cartridges with metal cases the value of h + 5s shall be equal to or less than 381mm (15"). For cartridge cases made of other materials, the value of h + 5s shall be established by the contractor prior to completion of Qualification testing.

3.4.1.2 Residual stress. The cartridge shall not split when subjected to the accelerated stress test for the applicable materials.

3.4.1.3 Bullet extraction. The force required to separate the bullet from the cartridge case shall not be less than 712 Newtons (160 pounds force).

3.4.1.4 Waterproof. The cartridge shall release not more than one bubble of air when tested at a pressure of 14 kPa (2 psi) below atmospheric pressure for 30 seconds. As an alternate procedure or for retesting the waterproof (firing) requirement specified in 3.4.3.2 may be applied.

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* 3.4.2 Chamber pressure. Using a government approved piezoelectric transducer, measuring at the midcase or case mouth position, the mean chamber pressure plus three standard deviations obtained with cartridges conditioned at 20 C (68 F) shall not exceed 200 MPa (29,000 psig).

3.4.2.1 Chamber pressure at extreme temperatures. The mean chamber pressure of cartridges conditioned at 50C (122F) and -20C (-4F) shall not increase more than 40 MPa (5,800 psi) above the mean chamber pressure obtained at 20C. The mean chamber pressure of cartridges conditioned at 50C (122F) and -20C (-4F) shall not exceed 200 MPa (29,000 psig). Decreased chamber pressure is acceptable.

3.4.3 Velocity. The mean velocity of cartridges conditioned at 20C (68F) measured at 24 m (78 ft) from the muzzle of the weapon shall not vary more than ± 5% from the design velocity. The design velocity is the mean velocity obtained during qualification testing at 20C.

3.4.3.1 Velocity at extreme temperatures. The mean velocity of cartridges conditioned at 50C (122F) and -20C (-4F) shall not vary by more than ± 8% from the design velocity.

3.4.3.2 Waterproof (firing). After immersion in 20C (68F) water for 24 hours the mean velocity of the test cartridges shall not vary from the mean velocity obtained in the 20C dry test by more than ± 30 mps (98fps).

3.4.4 Action time. The action time (overall primer ignition, propellant burning, plus the time taken for the bullet to exit the barrel) of cartridges conditioned at 20C (68F), 50C (122F) and -20C (- 4F) shall not exceed 4 milliseconds (ms).

3.4.5 Dispersion. When fired from the M2 BMG with M3 RA, the average mean radii of bullet marks on the targets shall not exceed 305mm (12 inches) at 150 meters.

3.4.5.1 Match. The mean point of impact of the tracer SRTA shall not be more than 150mm (5.9 inches) in the vertical direction from the mean point of impact of the ball SRTA when fired at targets located 150 meters from the M2 BMG with M3 RA.

3.4.6 Function and casualty. The cartridges shall function without casualty in M2 BMG with M3 RA throughout a temperature range of -20C to +50C. The cartridges shall safely function throughout a temperature range of -50C (-58F) through +70C (158F).

3.4.6.1 Cyclic rate. The cartridge shall function the M2 BMG with M3 RA between 450 and 600 shots per minute (spm) within the temperature range of -20C to +50C.

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3.4.6.2 Noise level. The noise levels of cartridges shall be no greater than 164 decibels nor more than 10 decibels below the noise level of M33 ball cartridges fired under identical conditions.

3.4.6.3 Breechflash. Breechflash when firing cartridges in the M2 BMG with M3 RA shall not be unsafe for the gunner.

3.4.6.4 Hot chamber effect. A cartridge inserted in the chamber immediately following the firing of 100 cartridges shall not fire (cook off). In addition the cartridge shall not deform to the extent that clearing of the weapon is made difficult following the proper waiting period.

3.4.6.5 Chemical compatibility. The reliability of the cartridge shall not be less than .900 when fired in the M2 BMG with M3 RA after having been immersed in the following lubricants and materials:

Specification	Material
MIL-L-46000	Lubricant, Semi-Fluid (LSA)
MIL-L-14107	Lubricating Oil, (LAW)
MIL-L-2104	Lubricating Oil, (ICE, T)
MIL-L-46167	Lubricating Oil, (ICE, A)
MIL-L-63460	Lubricant, (CPW)
VV-L-800	Lubricating Oil, (PL-S)
MIL-C-372	Bore Cleaner, (RBC)
O-I-503	Insect Repellant
O-T-620	Trichlorethane Solvent
P-D-680	Dry Cleaning Solvent (SD-1)

3.4.6.6 Belt pull. The cartridge shall function in the M2 BMG with M3 RA when a 100 round belt of cartridges is loaded in the weapon and allowed to hang freely.

3.4.6.7 Bullet integrity. The bullet of the cartridge shall not burst or disintegrate in the barrel or in flight when fired.

3.4.6.8 Ricochet. The bullet or bullet fragments shall not ricochet back more than 25 meters when fired at wood, metal, or masonry targets at a distance of 50 meters, nor outside the safety fan as shown in figure 1 when bullets are fired at level ground.

3.4.6.9 Muzzle flash. The muzzle flash made by firing cartridges in the M2 BMG with M3 RA shall be visible in darkness from 100 meters in front of the weapon.

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3.4.6.10 Fouling. The fouling produced by firing 1000 cartridges shall not cause more than one weapon stoppage, any safety problem, and shall be readily removable using standard cleaning methods. The fouling produced by firing 1000 cartridges shall not cause a change in cyclic rate from the first 100 rounds to the final 100 rounds greater than 15%. The fouling produced by firing 1000 cartridges shall not increase the dispersion more than 15% beyond the average mean radius obtained during initial dispersion tests.

3.4.6.11 Target kill. The test cartridges shall mark standard polyethylene and cardboard "E" and "F" silhouette targets sufficiently to activate M31A target mechanisms at ranges of 50, 100, and 150 meters with a reliability of .980.

3.4.6.12 Attitude. The cartridge shall operate the weapon when fired at an attitude of ± 85 degrees.

3.4.6.13 Maximum range. The maximum range of the bullet of the cartridge shall be 700 meters and shall be within the safety fan as shown on figure 1.

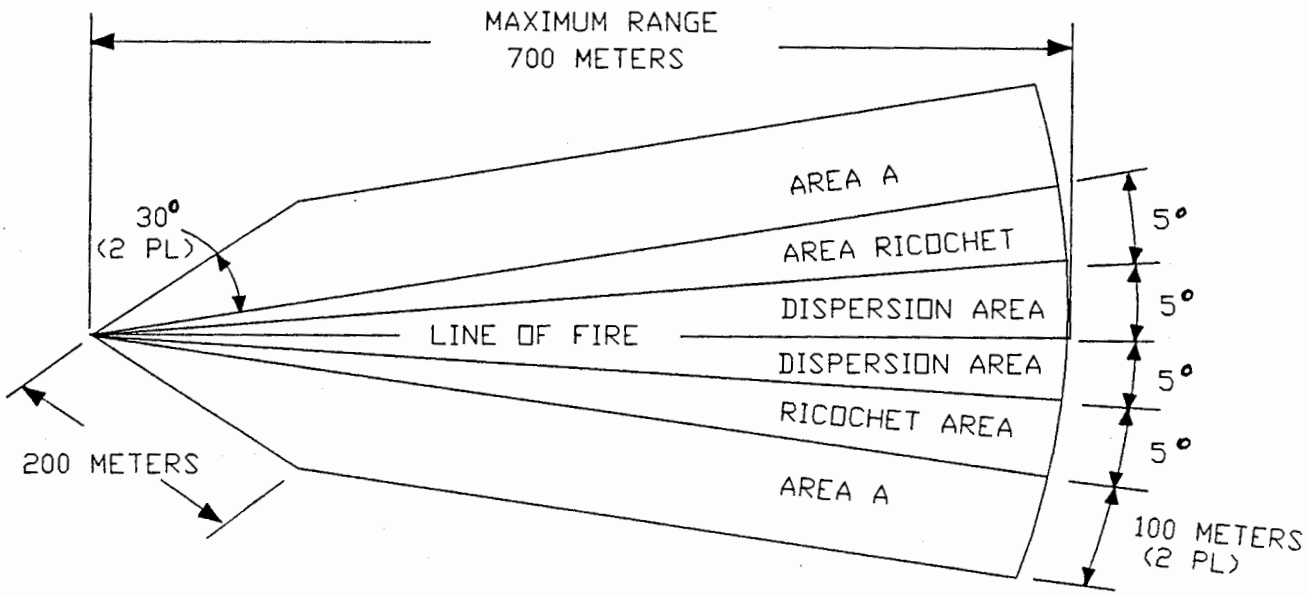


FIGURE 1. SAFETY FAN (SURFACE DANGER ZONE)

3.4.6.14 Trace. The bullet trace shall be visible at 20 meters to 150 meters from the weapon muzzle.

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3.5 Environmental.

3.5.1 Sand and dust. The reliability of the cartridge shall not be less than 90% of the reliability of the Caliber .50 M33 Ball cartridges when both are subjected to the sand and dust test.

3.5.2 Corrosion. The cartridge shall fire safely in the M2 BMG with M3 RA after being subjected to a 5% salt spray solution. The reliability of the cartridge shall be not less than .800 (90% confidence level).

3.5.3 Temperature/humidity cycling. After being subjected to a 10 day temperature/humidity cycle, the cartridge when fired in the M2 BMG with M3 RA shall not change the cyclic rate of the weapon by more than 15% and the reliability shall not be less than .900 (90% confidence level).

3.5.4 Long term high temperature. After being subject to 30 days of continuous heating at 65C (150F) and 10% to 20% relative humidity, the cartridge when fired in the M2 BMG with M3 RA shall not change the cyclic rate of the weapon by more than 20%. The reliability shall not be less than .900 (90% confidence level). If the cartridge fails to meet either of these requirements, the cartridge shall be retested after conditioning at 50C (122F) and 10% to 20% relative humidity for 120 days.

3.5.5 Long term low temperature. After being subject to 30 days of continuous cooling at -40C (-40F), the cartridge when fired in the M2 BMG with M3 RA shall not change the cyclic rate of the weapon by more than 20%. The reliability shall not be less than .900 (90% confidence level).

3.5.6 Rough handling.

3.5.6.1 Sequential. After being subjected to the transportation-vibration, loose cargo, 1.5 meter drop test, and the 2.1 meter drop tests; The cartridges shall be safe to handle, store, and fire. The reliability shall not be less than .900 (90% confidence level).

3.5.6.2 12.2 meter drop. After being subjected to the 12.2 meter drop test the cartridges shall not detonate, deflagrate, loose propellant, and must be safe to handle and to dispose.

3.5.7 Toxic fumes. The toxic fumes produced by the cartridge shall be acceptable to the Surgeon General for the applicable training scenarios.

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3.6 Safety.

3.6.1 Safety hazard classification. The safety hazard classification for the cartridge packed for shipping and storage shall be:

DOD HAZARD CLASS: 1.4
DOD STORAGE COMPATIBILITY GROUP: 5
DOD HAZARD CLASS: CLASS C EXPLOSIVE
DOD CONTAINER MARKING: SMALL ARMS AMMUNITION

3.6.2 Net explosive weight. The net explosive weight of the cartridge (sum of the propellant, primer, and any other pyrotechnic material shall not exceed 15 grams (230 grains).

3.6.3 Explosive sensitivity. The contractor shall provide explosive sensitivity data on all energetic material contained in the cartridges. The sensitivity for the materials shall be as indicated below:

Sensitivity	Requirement
a. Impact, ERL Type 12 tools 2.4 kg, cm @ 50% point	20 cm (min)
b. Friction Pendulum	
- Steel Shoe	Burn - no detonation
- Fiber Shoe	Burn - no detonation
c. Electrostatic Discharge .030 Joule	No detonation in 20 tests
d. Differential Thermal Analysis	200 C (min)

3.7 Workmanship. All plastic parts shall be fabricated in a thorough workmanlike manner. Plastic parts shall be free of porosity, twist, warpage, heat marks, sinks, voids, cracks, chipped edges, burrs, bubbles, blisters, burn marks, or other defects that will affect serviceability. All metal parts shall be fabricated in a thorough workmanlike manner. Metal parts shall be free of burrs, chips, chipped edges, gouges, cracks, splits, surface defects, dust, dirt, grease, oil, lacquer smears, and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by any cleaning agent. All assemblies shall be assembled in a thorough workmanlike manner. Extreme care should be exercised to avoid contamination of primers or propellant by oil, grease, or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification (See 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).

Note: Each cartridge configuration (Ball and Tracer) shall be subjected to all applicable inspections and tests specified herein.

4.3 Qualification.

4.3.1 Submission. The contractor shall submit a Qualification sample of Ball and Tracer Cartridges, and 500 Primed cases as designated by the Contracting Officer for evaluation IAW the provisions of 4.3.3.

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4.3.2 Qualification sample. The qualification sample shall be made to the same design as will be submitted for production. The qualification sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used for production. Where production equipment is not available, equipment that is representative of the production equipment shall be used. The qualification sample shall not be custom made in such a way that the product or product quality will be different from production ammunition.

* 4.3.3 Inspection to be performed. As determined by the Government, the qualification assemblies, components, and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (Table I), MIL-A-48078, and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.4 Rejection. See MIL-A-48078 and Tables I and II herein. Submission of a qualification sample that is not representative of normal production shall be cause for rejection of that sample and denial of design qualification. Failure of the qualification sample to be IAW the applicable drawings and the requirements of Tables I and II, as specified herein, will result in disapproval of the qualification sample and denial of qualification of the submitted design.

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TABLE I. Qualification.

CLASSIFICATION OF CHARACTERISTICS

Cartridge, Caliber .50 SRTA sh 1 of 3 12598606 12598607

CLASS	EXAMINATION OR TEST	CONFORMANCE CRITERIA AC	RE	REQUIREMENT PARAGRAPH	INSPECTION METHOD	REF
	Examination for defects					
	Critical	All	0 1	3.1	4.5.2.1	
	Major Class	800	2 3	3.1	4.5.2.1	
	Minor Class	800	10 11	3.1	4.5.2.1	
	Primer sensitivity	500	1/	3.4.1	4.6.1	
	Residual stress	50	0 1	3.4.1.2	4.6.2	
	Bullet extraction	50	0 1	3.4.1.3	4.6.3	
	Waterproof	50	0 1	3.4.1.4	4.6.4	
	Chamber pressure				4.6.5	2/
	a. 20C	50	1/	3.4.2		
	b. -20C	50	1/	3.4.2.1		
	c. 50C	50	1/	3.4.2.1		
	Velocity				4.6.6	2/
	a. 20C	50	1/	3.4.3		
	b. -20C	50	1/	3.4.3.1		
	c. 50C	50	1/	3.4.3.1		
	d. Waterproof	50	1/	3.4.3.2	4.6.6.1	
	Action time				4.6.7	2/
	a. 20C	50	0 1	3.4.4		
	b. -20C	50	0 1	3.4.4		
	c. 50C	50	0 1	3.4.4		
	Dispersion at 150 meters	300	1/	3.4.5	4.6.8	
	Match	300	1/	3.4.5.1	4.6.8.1	

1/ The design shall be rejected if the requirement is not met.
2/ These tests may be performed concurrently.

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TABLE I. Qualification.

CLASSIFICATION OF CHARACTERISTICS

Cartridge, Caliber .50 SRTA sh 2 of 3 12598606 12598607

CLASS	EXAMINATION OR TEST	CONFORMANCE CRITERIA AC	RE	REQUIREMENT PARAGRAPH	INSPECTION METHOD	REF
	Function & Casualty		3/	3.4.6	4.6.9	
	M2 BMG with M3 RA on rigid mount 4/					
	a. 20C	3000				
	b. -20C	3000				
	c. 50C	3000				
	d. -50C	200	5/			
	e. 70C	200	5/			
	M2 BMG with M3 RA on M3 tripod mount at -20C	1000				
	M2 BMG with M3 RA on M4 pedestal mount at -20C	1000				
	Cyclic rate on rigid mount	11,000	1/	3.4.6.1	4.6.10	
	Noise 6/	100	1/	3.4.6.2	4.6.11	
	Breechflash	7	0 1	3.4.6.3	4.6.12	
	Hot chamber	303	1/	3.4.6.4	4.6.13	
	Chemical compatibility	380	1/	3.4.6.5	4.6.14	
	Belt pull	100	1/	3.4.6.6	4.6.15	
	Bullet integrity	50	0 1	3.4.6.7	4.6.16	
	Ricochet	90	0 1	3.4.6.8	4.6.17	
	Muzzle flash	20	0 1	3.4.6.9	4.6.18	
	Fouling	1180	1/	3.4.6.10	4.6.19	

1/ The design shall be rejected if the requirement is not met.
2/ These tests may be performed concurrently.
3/ See Table II for accept/reject criteria.
4/ The weapon shall be conditioned at the same temperature as the ammunition.
5/ These tests for safety evaluation only. Any safety problem caused by the cartridge shall be cause for rejection.
6/ These tests may be performed concurrently with the function and casualty test.
7/ To be observed during all firing tests.

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TABLE I. Qualification.

CLASSIFICATION OF CHARACTERISTICS

Cartridge, Caliber .50 SRTA sh 3 of 3 12598606 12598607

CLASS	EXAMINATION OR TEST	CONFORMANCE CRITERIA AC	RE	REQUIREMENT PARAGRAPH	INSPECTION METHOD REF
	Target kill	300	6 7	3.4.6.11	4.6.20
	Attitude	20	0 1	3.4.6.12	4.6.21
	Maximum range	165	1/	3.4.6.13	4.6.22
	Trace	100	10 11	3.4.6.14	4.6.23
	Sand and dust	100	1/	3.5.1	4.6.24
	Corrosion	50	1/	3.5.2	4.6.25
	Temperature/humidity cycling	92	5 6	3.5.3	4.6.26
	Long term high temperature	92	5 6	3.5.4	4.6.27
	Long term low temperature	92	5 6	3.5.5	4.6.28
	Rough handling, sequential	400	32 33	3.5.6	4.6.29.1
	Rough handling, 12.2 meter drop	400	1/	3.5.6.2	4.6.29.2
	Toxic fumes	300	1/	3.5.7	4.6.30
	Safety hazard classification		1/	3.6	4.6.31
	Trace	100	10 11	3.4.6.14	4.6.23

1/ The design shall be rejected if the requirement is not met.

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TABLE II. Qualification inspection - function and casualty firing defect acceptance - rejection criteria.

	Defect	Acceptance Number 1/	Rejection Number
1.	Bullet in bore	0	1
2.	Burn through	0	1
3.	Misfire - no vent hole 2/	0	1
4.	Weapon stoppage attributable to ammunition	11	12
5.	Separation of case base from body (if applicable)	2	3
6.	Failure to extract	1	2
7.	Misfire excluding no vent	2	3
8.	Primer defects:		
	a. Perforation in the primer cup caused by the firing pin	4/	4/
	b. Perforation in the primer cup not caused by the firing pin	5/	5/
	c. Escape of gas around the primer cup	4/	4/
	d. Blown primer	3	4
	e. Primer that fall out of pocket on retraction of the bolt	3	4
9.	Case casualties 6/		
	a. Longitudinal split		
	(1) Body (K) extending to case base	5/	5/
	(2) Case base (L)	5/	5/
	(3) Through case base (M)	5/	5/
	(4) Upper body (J)	4/	4/
	b. Circumferential rupture		
	(1) Partial - neck, shoulder, or body (S, J, or K)	5/	5/
	(2) Partial - case base (L)	5/	5/
	(3) Complete	0	1

NOTES:

1/ If the function and casualty defects plus firing defects observed in all other tests exceed the acceptance number or ratio specified, the qualification sample shall be rejected.

2/ Each cartridge that misfires shall be disassembled and examined for a blocked vent hole.

3/ All stoppages attributable to the ammunition, with the exception of misfire, complete circumferential rupture, or failure to extract, observed in all tests shall be included. Occasionally the last round of a continuous belt, due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure shall not be classified as a weapon stoppage nor shall it be recorded as a defect chargeable to the ammunition.

4/ The defect rate for all these defects combined shall not exceed 67 per 1000 rounds.

5/ The defect rate for all these defects combined shall not exceed 20 per 1000 rounds.

6/ For location of defects indicated in parenthesis, see drawing 7643674.

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4.4 First article inspection.

4.4.1 First article submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation IAW provisions of Table IV. The first article sample shall consist of the assemblies, components, and test specimens listed below in the quantities indicated:

Part Description	Drawing	Quantity
Cartridge, Cal.50 Ball SRTA	12598607	4,000 cartridges and 500 primed cases
Cartridge, Cal.50 Tracer SRTA	12598606	4,000 cartridges and 500 primed cases

4.4.2 Inspection to be performed. As determined by the Government, the first article assemblies, components, and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (Table IV), MIL-A-48078, and be inspected for compliance with any or all requirements of the applicable drawings.

4.4.3 Rejection. See MIL-A-48078. Failure of the First Article sample to be IAW the applicable drawings and requirements of Table III. and Table IV. as specified herein, will result in disapproval of the First Article.

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TABLE III. First Article Inspection - function and casualty firing defect acceptance - rejection criteria.

	Defect	Acceptance Number	Rejection Number
1.	Bullet in bore	0	1
2.	Burn through	0	1
3.	Misfire - no vent hole	0	1
4.	Weapon stoppage attributable to ammunition 1/	3	4
5.	Separation of case base from body (if applicable)	0	1
6.	Failure to extract	0	1
7.	Misfire excluding no vent	1	2
8.	Primer defects:		
	a. Perforation in the primer cup caused by the firing pin	24	25
	b. Perforation in the primer cup not caused by the firing pin	7	8
	c. Escape of gas around the primer cup	14	15
	d. Blown primer	0	1
	e. Primer that fall out of pocket on retraction of the bolt	0	1
9.	Case casualties 2/		
	a. Longitudinal split		
	(1) Body (K) extending to case base	8	9
	(2) Case base (L)	0	1
	(3) Through case base (M)	0	1
	(4) Upper body (J)	14	15
	b. Circumferential rupture		
	(1) Partial - neck, shoulder, or body (S, J, or K)	1	2
	(2) Partial - case base (L)	0	1
	(3) Complete	0	1

NOTES:

1/ Occasionally the last round of a continuous belt, due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure shall not be classified as a weapon stoppage nor shall it be recorded as a defect chargeable to the ammunition. All other stoppages attributable to the ammunition with the exception of misfire, failure to extract and complete circumferential rupture shall be included in this defect category.

2/ For location of defects indicated in parenthesis, see drawing 7643674.

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TABLE IV. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

Cartridge, Caliber .50 SRTA — sh 1 of 2 — 12598606 12598607

CLASS	EXAMINATION OR TEST	CONFORMANCE CRITERIA AC	RE	REQUIREMENT PARAGRAPH	INSPECTION METHOD REF
	Examination for defects				
	Critical	All	0 1	3.1	4.5.2.1
	Major Class	800	7 8	3.1	4.5.2.1
	Minor Class	800	9 10	3.1	4.5.2.1
	Primer sensitivity	500	1/	3.4.1	4.6.1
	Residual stress	50	0 1	3.4.1.2	4.6.2
	Bullet extraction	50	0 1	3.4.1.3	4.6.3
	Waterproof	50	0 1	3.4.1.4	4.6.4
	Chamber pressure				4.6.5
	a. 20C 2/	50	1/	3.4.2	
	b. -20C 2/	50	1/	3.4.2.1	
	c. 50C 2/	50	1/	3.4.2.1	
	Velocity				4.6.6
	a. 20C 2/	50	1/	3.4.3	
	b. -20C 2/	50	1/	3.4.3.1	
	c. 50C 2/	50	1/	3.4.3.1	
	Waterproof	50	1/	3.4.3.2	4.6.6.1
	Action time				4.6.7
	a. 20C 2/	50	0 1	3.4.4	
	b. -20C 2/	50	0 1	3.4.4	
	c. 50C 2/	50	0 1	3.4.4	

1/ The sample shall be rejected if the requirement is not met.
2/ These tests may be performed concurrently.

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TABLE IV. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

Cartridge, Caliber .50 SRTA sh 2 of 2 12598606 12598607

CLASS	EXAMINATION OR TEST	CONFORMANCE CRITERIA	AC	RE	REQUIREMENT PARAGRAPH	INSPECTION METHOD REF
	Dispersion at 150 meters 2/	300	1/		3.4.5	4.6.8
	Match 2/	180	1/		3.4.5	4.6.8.1
	Function & Casualty		3/		3.4.6	4.6.9
	M2 BMG with M3 RA on rigid mount 2/					
	a. 20C	900				
	b. -20C	900				
	c. 50C	900				
	Cyclic rate on rigid mount 4/	900	1/		3.4.6.1	4.6.10
	Breechflash	5/	0	1	3.4.6.3	4.6.12
	Trace	100	10	11	3.4.6.14	4.6.23

1/ The sample shall be rejected if the requirement is not met.
2/ These tests may be performed concurrently.
3/ See Table II for accept/reject criteria.
4/ These tests may be performed concurrently with the function and casualty test.
5/ To be observed during all Function & Casualty tests.

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4.5 Quality conformance inspection.

4.5.1 Inspection lot formation. IAW MIL-A-48078.

4.5.1.1 Component parts. Unless otherwise specified, component parts shall be homogeneous and of a size convenient to the contractor and inspected, tested, and accepted by the contractor. The cartridge lot shall contain:

a. Cartridge cases from one manufacturer.
b. Penetrators from one unchanged process and one manufacturer.
c. Primers from one interfix and not more than 2 lots.
d. Propellant from one interfix and not more than 2 lots.

4.5.1.2 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, lot number IAW MIL-STD-1168, and the supplier's identification as assigned by the procuring activity.

4.5.1.3 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.5.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.5.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. Attributes sampling inspection shall be conducted IAW Table V using the inspection levels cited in the Classification of Characteristics paragraphs.

Critical defects. Unless otherwise specified, 100% examination shall be performed for all critical defects. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot shall then be screened 100% for the critical defect and resubmitted for acceptance.

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* TABLE V. Attributes sample inspection.

	INSPECTION LEVELS	
	III	IV
LOT SIZE	SAMPLE SIZE	
2 TO 8	*	*
9 TO 15	*	13
16 TO 25	*	13
26 TO 50	32	13
51 TO 90	32	13
91 TO 150	32	13
151 TO 280	32	32
281 TO 500	50	32
501 TO 1200	80	50
1201 TO 3200	80	50
3201 TO 10000	125	50
10001 TO 35000	125	80
35001 TO 150000	125	80
150001 TO 500000	200	125
500001 AND OVER	200	125

Asterisks (*) indicate that 100% inspection shall be performed. If the sample size exceeds the lot size, perform 100% inspection. Accept with zero defects only and reject with one or more defects for all inspection levels.

b. Alternative quality conformance. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling, or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.8). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provide equivalent assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

4.5.2.1 Cartridge, Caliber .50 SRTA sh 1 of 2 12598606 12598607

CLASS	EXAMINATION OR TEST 1/	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REF
CRITICAL:				
1.	CASE SPLIT IN K, L, OR M LOCATION (6)	100%	3.1	GAGE
2.	CASE SPLIT IN I, S, OR J LOCATION (6) WHEN LOSS OF POWDER OCCURS	100%	3.1	GAGE
3.	PERFORATED CASE (7)	100%	3.1	GAGE
4.	WEIGHT UNDER MINIMUM 2/	100%	3.1	GAGE
SPECIAL:				
a.	PRIMER MISSING (32)	100%	3.1	VISUAL
b.	PRIMER COCKED (33)	100%	3.1	VISUAL
c.	PRIMER INVERTED (34)	100%	3.1	VISUAL
MAJOR:				
101.	CASE SPLIT IN I, S, OR J LOCATION (6) WITH NO LOSS OF POWDER	LEVEL III	3.1	VISUAL
102.	CORRODED OR STAINED (IF ETCHED) CASE (2)	LEVEL III	3.1	VISUAL
103.	ROUND HEAD (CASE) (4)	LEVEL III	3.1	VISUAL
104.	DENTED CASE (5)	LEVEL III	3.1	VISUAL
105.	DRAW SCRATCH IN CASE (8)	LEVEL III	3.1	VISUAL
106.	BEVELED UNDERSIDE OF HEAD (10)	LEVEL III	3.1	VISUAL
107.	SCALY METAL (CASE) (12)	LEVEL III	3.1	VISUAL
108.	NO CHAMFER ON HEAD (RIM) (13)	LEVEL III	3.1	VISUAL
109.	LOOSE PRIMER (35)	LEVEL III	3.1	VISUAL/ MANUAL
110.	CARTRIDGE PROFILE FAILURE 3/	LEVEL III	3.1	GAGE
111.	DIAMETER OF EXTRACTOR GROOVE, MAXIMUM	LEVEL III	3.1	GAGE
112.	DIAMETER OF HEAD, MAXIMUM	LEVEL III	3.1	GAGE
113.	THICKNESS OF RIM, MAXIMUM	LEVEL III	3.1	GAGE
114.	DISTANCE TO SHOULDER DATUM	LEVEL III	3.1	GAGE
115.	DEPTH OF PRIMER	LEVEL III	3.1	GAGE
116.	OVERALL LENGTH, MAXIMUM	LEVEL III	3.1	GAGE

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

4.5.2.1 Cartridge, Caliber .50 SRTA sh 2 of 2 12598606 12598607

CLASS	EXAMINATION OR TEST 1/	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REF
MINOR:				
201.	DISCOLORED, DIRTY, OILY, OR SMEARED (WATERPROOFING)	LEVEL IV	3.1	VISUAL
202.	DENTED CASE (5)	LEVEL IV	3.1	VISUAL
203.	DRAW SCRATCH IN CASE (8)	LEVEL IV	3.1	VISUAL
204.	SCRATCH IN CASE (9)	LEVEL IV	3.1	VISUAL
205.	SCALY METAL ON CASE (12)	LEVEL IV	3.1	VISUAL
206.	FOLD, WRINKLE BUCKLE, OR BULGE IN CASE (14, 15, 16, 17)	LEVEL IV	3.1	VISUAL
207.	HEAD STAMP MISSING OR ILLEGIBLE (18)	LEVEL IV	3.1	VISUAL
208.	DEFECTIVE HEAD (19)	LEVEL IV	3.1	VISUAL
209.	NO WATERPROOFING MATERIAL (PRIMER POCKET JOINT) (37), IF APPLICABLE	LEVEL IV	3.1	VISUAL
210.	DEFECTIVE CRIMP (PRIMER (38) IF APPLICABLE	LEVEL IV	3.1	VISUAL
211.	DIAMETER OF EXTRACTOR GROOVE, MINIMUM	LEVEL IV	3.1	GAGE
212.	WORKMANSHIP	LEVEL IV	3.1	VISUAL

NOTES:

1/ Numbers after defect description refer to visual standards in MIL-STD-636 (caliber .50 section). While no photographs of SRTA cartridges are included in this standard, the cartridge case defects and primer defects illustrated in the caliber .50 section can be used. In the event of conflict between 4.5.2.1 of this specification and MIL-STD-636 as to defect classification, the classification specified in 4.5.2.1 shall apply.

2/ The contractor shall submit an analysis that establishes the minimum charge required to preclude the possibility of a bullet in bore. Cartridges containing less than this minimum charge are considered to be "critical defects".

3/ Requiring more than 80 pound dead weight to insert in the profile and alignment gage.

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4.5.3 Testing. Testing is described in 4.5.3.1. Testing shall be conducted IAW the methods and procedures specified in 4.5.

a. Test samples. The quantities for the various tests shall be as specified in 4.5.3.1. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. Sufficient cartridges shall be selected so that all testing (including testing of second samples where necessary) can be performed. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

b. Firing defects. Firing defects and associated acceptance-rejection criteria are specified in Table VI.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

4.5.3.1 Cartridge, Caliber .50 SRTA sh 1 of 2 12598606 12598607

CLASS	EXAMINATION OR TEST	CONFORMANCE CRITERIA	AC	RE	REQUIREMENT PARAGRAPH	INSPECTION METHOD	REF
Primer sensitivity		500	1/		3.4.1	4.6.1	
Residual stress		50/50	2/		3.4.1.2	4.6.2	
Bullet extraction		50/50	2/		3.4.1.3	4.6.3	
Waterproof		50/50	2/		3.4.1.4	4.6.4	
	Chamber pressure at 20C 4/	50/100	3/		3.4.2	4.6.5	
	Velocity						
	a. 20C 4/	50/100	3/		3.4.3	4.6.6	
	b. Waterproof at 20C	50/100	3/		3.4.3.2	4.6.6.1	
	Action time at 20C 4/	50/100	3/		3.4.4	4.6.7	
	Dispersion	90/180	3/		3.4.5	4.6.8	
	Function & Casualty M2 BMG with M3 RA on a rigid mount		5/		3.4.6	4.6.9	
	a. 20C	300/600					
	b. -20C	300/600					
	c. 50C	300/600					
	Cyclic rate on rigid mount 6/	300/600			3.4.6.1	4.6.10	
	Breechflash	7/	0	1	3.4.6.3	4.6.12	
	Trace	100	10	11	3.4.6.14	4.6.23	

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4.5.3.1 QUALITY CONFORMANCE INSPECTION

notes:

1/ The lot shall be rejected if the requirement is not met.

2/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test a second sample consisting of an equal number of cartridges shall be tested. If any additional cartridges fail in the second test the lot shall be rejected.

3/ Failure of the cartridges to comply with the applicable requirement shall result in rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of cartridges in the second sample to comply with the applicable requirement shall result in rejection of the lot.

4/ These tests may be performed concurrently.

5/ See Table VI.

6/ The cyclic rate shall be measured during the function and casualty test. Failure of any 100 round burst to comply wit the applicable requirement shall be cause for rejection of the lot, subject to testing a second sample consisting of an equal number of cartridges.

7/ To be observed during all Function & Casualty tests.

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4.5.4 Function and casualty. Firing defects and associated acceptance-rejection criteria are specified in Table VI.

Table VI. Quality Conformance Inspection - function and casualty firing defect classification and acceptance - rejection criteria

DEFECT	ACCEPTANCE NUMBERS 1/ FIRST SAMPLE	CUMULATIVE SAMPLE	DEFECT CLASSIFICATION
1. Bullet in bore	0	-	Critical
2. Burn through	0	-	Critical
3. Misfire - no vent hole	0	-	Major
4. Weapon stoppage attributable to ammunition 2/	3	6	Major
5. Separation of case base from body (if applicable)	0	1	Major
6. Failure to extract	0	1	Major
7. Misfire excluding no vent	1	2	Major
8. Primer defects:			
a. Perforation in the primer cup caused by the firing pin	18	45	Minor
b. Perforation in the primer cup not caused by the firing pin	6	13	Major
c. Escape of gas around the primer cup	12	25	Minor
d. Blown primer	0	1	Major
e. Primer that falls out of pocket on retraction of the bolt	0	1	Major
9. Case casualties 3/			
a. Longitudinal split			
(1) Body (K) extending to case base	7	15	Major
(2) Case base (L)	0	1	Major
(3) Through case base (M)	0	1	Major
(4) Upper body (J)	12	25	Minor
b. Circumferential rupture			
(1) Partial - neck, shoulder, or body (S, J, or K)	2	4	Major
(2) Partial-case base (L)	0	1	Major
(3) Complete	0	1	Major

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Table VI. (Cont.)

NOTES:

1/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other tests exceed the acceptance number for the cumulative sample. If the number of defects found in the first test exceeds the acceptance number, a second sample, consisting of double the quantities specified for the function and casualty test, shall be fired. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that the acceptance number for the cumulative sample is exceeded, the lot shall be rejected. If no acceptance number is listed for the cumulative sample, no second sample shall be permitted.

2/ Occasionally the last round of a continuous belt, due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure shall not be classified as a weapon stoppage nor shall it be recorded as a defect chargeable to the ammunition. All other stoppages attributable to the ammunition with the exception of misfire, failure to extract and complete circumferential rupture shall be included in this defect category.

3/ For location of defects indicated in parenthesis, see drawing 7643674.

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4.5.5 Packaging, packing, and marking inspection. Inspection for linking, packaging, packing and marking shall be IAW MIL-STD-644 as applies to the drawing or as required by the contract.

4.5.6 Inspection equipment. The inspection equipment required to perform the examination and test prescribed herein is described in the Inspection Method Reference column in the Qualification tables, First Article examination tables, and Quality Conformance Inspection Classification of Characteristics paragraphs. The contractor shall submit inspection equipment designs for approval by the Government IAW the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5.7 Inspection equipment lists. The examination and tests shall be made using equipment listed on LI12598606 and LI12598607 or Government approved equipment.

* 4.6 Methods of inspection. All firing tests excepting chamber pressure, velocity, velocity/ waterproof, and action time tests are to be conducted using belted ammunition consisting of cartridges with M9 links in the quantities indicated for each test and linked in a manner appropriate for the test.

4.6.1 Primer sensitivity. Test IAW the procedure specified in Section 7-25 of TECP 700-700 Vol III or Government approved procedures.

4.6.2 Residual stress.

4.6.2.1 Cartridges with brass cases. Test IAW paragraph 14 of Section 7-24 of TECP 700-700 Vol III.

4.6.2.2 Cartridges with steel cases. Testing not required.

4.6.2.3 Cases of other materials. Test IAW procedures approved by the Government appropriate to the material(s) used to manufacture the cartridge.

4.6.3 Bullet extraction. The cartridge shall be tested in an approved bullet extraction machine. The rate of travel of the test head shall be from 75mm per minute to 150mm per minute (3 to 6 inches per minute).

4.6.4 Waterproof. The cartridge shall be placed horizontally in a clear bell jar and covered with at least 26mm (1 inch) of water. The jar shall be evacuated to the specified differential pressure. Observation for air bubbles escaping from the cartridges shall be performed for thirty seconds minimum.

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* 4.6.5 Chamber pressure. Cartridges conditioned at 20C ±2C (68F ±3F) for not less than two hours, -20C ±5C (-4F ±9F) for not less than six hours, or -50C±5C (-58F ±9F) and 50C ±5C (122F ±9F) for four to twelve hours shall be fired for chamber pressure measurements one shot at a time in a universal receiver instrumented with a piezoelectric transducer installed at the midcase or case mouth position. The chamber pressure test shall be conducted IAW SCATP 5.56, chapter 1, Section 18. Note: The port pressure portion of the test does not apply to the caliber .50 SRTA cartridges. The use of test barrel, 8649459, is mandatory.

4.6.6 Velocity. Cartridges conditioned at 20C ±2C (68F ±3F) for not less than two hours, -20C ±5C (-4F ±9F) for not less than six hours, and 50C ±5C (122F ±9F) for four to twelve hours shall be fired one shot at a time for velocity measurements in a universal rifle. The test shall be conducted IAW TECP 700-700, Vol. III, Section 7-3. The use of test barrel, 8649459, is mandatory.

4.6.6.1 Velocity/waterproof. The cartridges shall be submerged in water for 24 hours, wiped dry, then placed in a temperature controlled room or chamber at 20C ±2C (68F ±3F) for not less than two hours. The cartridges shall be fired one shot at a time for velocity measurements in a universal rifle. The test shall be conducted IAW TECP 700-700, Vol. III, Section 7-3. The use of test barrel, 8649459, is mandatory.

Note: The contractor shall produce and make available a reference lot of SRTA cartridges. The procedures for assessing chamber pressure and velocity shall be as specified in MIL-C-14858.

4.6.7 Action time. Cartridges conditioned at 20C ±2C (68F ±3F) for not less than two hours, -20C ±5C (-4F ±9F) for not less than six hours, and 50C ±5C (122F ±9F) for four to twelve hours shall be fired one shot at a time for action time measurements in a universal rifle instrumented for recording action time. The test shall be conducted IAW SCATP 7.62 , Section 4. The use of test barrel, 8649459, is mandatory.

4.6.8 Dispersion. The cartridges shall be tested in an M2 BMG with M3 RA which is attached to a rigid mount. The test shall be conducted IAW TECP 700-700, Vol. III, Section 7-14. Ten shots shall be fired at each target.

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4.6.8.1 Match. The cartridges shall be tested in an M2 BMG with M3 RA which is attached to a rigid mount. The target holder shall be designed to permit the use of registration marks to permit the removal of the target holder between the firing of the ball and tracer SRTA to show the location of the tracer SRTA dispersion relative to the ball SRTA dispersion. Once properly aimed at the target the weapon shall not be moved during the test. Fire 10 ball SRTA test cartridges IAW TECP 700-700, Vol. III, Section 7-14. Mark the holes. Without moving the weapon fire 10 tracer SRTA test cartridges IAW TECP 700-700, Vol. III, Section 7-14. Repeat the firing sequence until the required number of test targets have been fired. Measure and record the vertical distance between the mean points of impact of the ball and tracer bullet holes on each target.

4.6.9 Function and casualty. The cartridges shall be tested in an M2 BMG with M3 RA which is attached to the specified mount (M3 tripod, M4 pedestal or hard mount). The cartridges shall be conditioned at 20C ±2C (68F ±3F) for not less than two hours, -20C ±5C (-4F ±9F) for not less than six hours, and 50C ±5C (122F ±9F) for four to twelve hours, -50C ±5C (-58F ±9F) for not less than six hours, and 70C ±5C (158F ±9F) for four to six hours. The weapons shall be at room temperature at the beginning of the test and shall be cooled between bursts. The function and casualty test shall be conducted IAW TECP 700-700, Vol. III, Section 7-15. The test ammunition shall be linked in 300 round belts. After an initial 100 round burst during which the cyclic rate is recorded, fire the remaining ammunition in 10 round bursts. Ammunition links shall not be reused in this test. Observe for compliance with Table II, Table III, or Table VI, as applicable.

4.6.10 Cyclic rate. Measure and record the cyclic rate during function and casualty testing at 20C (68F).

4.6.11 Noise. Fire ten ten round bursts of M33 Ball and ten ten round bursts of test cartridges placing the microphone of the noise level meter approximately .5 meters directly behind the weapon.

4.6.12 Breechflash. Observe breechflash during function and casualty testing from a safe vantage point. Record any breechflashes observed and any breechflash that may be unsafe for the gunner.

4.6.13 Hot chamber. Using an M2 BMG with M3 RA fire 100 continuous rounds and immediately bring the test cartridge into battery. Record the time if the test round fires. After 25 minutes remove the test cartridge and examine and record any damage to the test cartridge or weapon barrel chamber. Repeat two times.

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4.6.14 Chemical compatibility. Thirty eight cartridges each shall be dipped in the following compounds and then drained for 24 hours:

	Specification	Material
a.	MIL-L-46000	Lubricant, Semi-Fluid (LSA)
b.	MIL-L-14107	Lubricating Oil, (LAW)
c.	MIL-L-2104	Lubricating Oil, (ICE, T)
d.	MIL-L-46167	Lubricating Oil, (ICE, A)
e.	MIL-L-63460	Lubricant, (CPW)
f.	VV-L-800	Lubricating Oil, (PL-S)
g.	MIL-C-372	Bore Cleaner, (RBC)
h.	O-I-503	Insect Repellant
i.	O-T-620	Trichlorethane Solvent
j.	P-D-680	Dry Cleaning Solvent (SD-1)

The test cartridges shall then be wiped dry and fired in ten round bursts in an M2 BMG with M3 RA IAW TECP 700-700, Vol. III, Section 7-15. Record the number of failures.

4.6.15 Belt pull. The cartridges shall be fired in ten round bursts from an M2 BMG with M3 RA with a 100 round belt of cartridges allowed to hang freely. The test shall be conducted IAW TECP 700-700, Vol. III, Section 7-15.

4.6.16 Bullet integrity. Using an M2 BMG with M3 RA test IAW TECP 700-700, Vol. III, Section 7-24, para 1, firing one shot at a time.

4.6.17 Ricochet. Using an M2 BMG with M3 RA, the test cartridges shall be fired one shot at a time at the following targets:

a. Plywood - 25mm thick x 1 meter minimum wide x 2 meters high. Place the plywood 50 meters from the muzzle of the weapon. Fire ten test cartridges.

b. Concrete block wall - 1 meter minimum wide x 2 meters minimum high. Place the concrete block wall 50 meters from the muzzle of the weapon. Fire ten test cartridges.

c. Steel plate - 10mm (.039 inch) thick x 1 meter minimum wide x 2 meters minimum high. Place the steel plate 50 meters from the muzzle of the weapon. Fire thirty test cartridges.

d. Level ground - Fire ten cartridges each at 2 degrees, 3 degrees, 4 degrees, and 6 degrees downward from the horizontal.

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4.6.18 Muzzle flash. The test shall be conducted IAW TECP 700-700, Vol. III, Section 7-24, paragraph 6, using an M2 BMG with M3 RA firing ten round bursts.

4.6.19 Fouling. The test shall be conducted as follows:

a. Fire 90 rounds of SRTA for dispersion (see 4.6.8) at 150m IAW TECP 700-700, Vol. III, Section 7-15, firing one shot at a time.

b. Continuing on with the same M2 BMG with M3 RA fire 1000 rounds of SRTA recording the cyclic rate of the initial 100 rounds and the final 100 rounds fired, firing the intervening 800 rounds in ten round bursts. This part of the test may be conducted concurrently with the function and casualty testing at 20C.

c. Upon completion of the 1000 round fouling test, fire 90 rounds of SRTA for dispersion (see 4.6.8) without having cleaned the weapon. Fire one shot at a time.

4.6.20 Target kill. The test shall be conducted as follows: Twenty five rounds shall be fired at each of the target types at ranges of 50, 100, 150 meters. The targets shall be polyethylene and cardboard "E" and "F" silhouette targets and shall be mounted on the M31A1 target holding mechanisms. A witness screen with minimum dimensions of 3m (10 ft) x 3m (10 ft) shall be positioned behind the targets. The cartridges shall be fired one shot at a time from the M2 BMG with M3 RA. An observer shall note if the target mechanism was activated and the results shall be recorded. If a shot does not activate a target mechanism the witness screen shall be examined to determine if the bullet struck the target. If there is a mark or hole in the witness screen, the hole shall be marked to distinguish the hole from subsequent holes. If it determined that a bullet did not strike the target mechanism, another nonpenalty shot shall be fired. Each shot and the result shall be recorded.

4.6.21 Attitude. Ten cartridges shall fired from an M2 BMG with M3 RA at an elevation of +85 degrees from the horizontal. Ten cartridges shall fired from an M2 BMG with M3 RA at an elevation of -85 degrees from the horizontal.

4.6.22 Maximum range. Using an M2 BMG with M3 RA the test shall be conducted at elevations between 0 and 45 degrees. Five shots shall be fired, one shot at a time, at each 5 degree increment and a determination shall be made as to what 10 degree elevation band provides the maximum range. Once this band has been established, ten additional rounds shall be fired at each 1 degree increment within the band and the maximum range shall be determined.

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4.6.23 Trace. Using an M2 BMG with M3 RA test IAW TECP 700-700, Vol. III, Section 7-18. Observation for trace performance shall be made at the weapon and at points 20 meters and 150 meters beyond the muzzle of the weapon on a line parallel to and approximately 70 meters (75 yards) from the line of trajectory or from a suitably placed observation tower. Visible markers shall be placed opposite the 20 and 150 meter observation points to insure a line of sight from each observation point perpendicular to the plane of the trajectory. The ammunition shall not be conditioned to temperature. Fire at least three warmer shots to sight, warm, and foul the weapon. Then fire the sample cartridges in regular sequence allowing sufficient time between shots for each observer to record trace results at each observation point. The date and each defect shall be recorded. Defects reported at more than one observation point for the same shot shall be recorded as a single failure. During firing observation shall also be made for (a) bullet bursting before striking bullet stop or target, (b) trace muzzle flash, and (c) erratic flight.

4.6.24 Sand and dust. The M2 BMG with M3 RA assembled with a 100 round cartridge belt shall be placed in the sand and dirt chamber. A sand and dust mixture as specified in TOP-3-2-045, Table 12, shall be dispersed on the weapon at a rate of 6.5 grams ± 2 grams per minute per square meter (100 ± 25 grains per minute per square meter) over the area concerned. The firing cycle shall begin one minute after the dispenser is turned on. The weapon shall be fired in 10 round bursts with a one minute interval between each burst. The test shall be repeated with Caliber .50 M33 Ball cartridges using the M2 BMG with the M3 RA removed. Cartridge failures shall be recorded.

* 4.6.25 Corrosion. The cartridge shall be subjected to a 5% salt spray solution for 48 hours. The test shall be performed IAW MIL-STD-810, Test 509. After completion of the test the cartridge shall be chambered and fired safely in the M2 BMG with M3 RA. Observation shall be made for compliance with the applicable requirement.

* 4.6.26 Temperature/humidity cycling. The cartridges shall be conditioned IAW MIL-STD-810, Test 507, Procedure III. After completion of the conditioning the cartridges shall be fired in 10 round bursts in an M2 BMG with M3 RA. The cyclic rate shall be recorded.

4.6.27 Long term high temperature storage. After storage for 30 days of continuous heating at 65C ±5C (150F ±9F) and 10% to 20% relative humidity, condition the cartridges at 20C ±2C (68F ±3F) for 24 hours. Upon completion of the conditioning the cartridges shall be fired in 10 round bursts in an M2 BMG with M3 RA. The cyclic rate shall be recorded and any cartridge failures shall be recorded.

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If any cartridges fail the test at 65C, a retest shall be performed on cartridges stored at 50C ±5C (122F ±3F) and 10% to 20% relative humidity for 120 days.

4.6.28 Long term low temperature storage. After storage for 30 days at -40C ±5C (-40F ±9F), condition the cartridges at 20C ±2C (68F ±3F) for 24 hours. Upon completion of the conditioning the cartridges shall be fired in 10 round bursts in an M2 BMG with M3 RA. The cyclic rate shall be recorded and any cartridge failures shall be recorded.

4.6.29 Rough handling.

4.6.29.1 Transportation-vibration and drop tests. Four hundred cartridges packed IAW paragraph 5.2.1 shall be subjected to the transportation-vibration test IAW TOP 1-2-601. Two hundred rounds shall be tested at -50C ±5C (-58F ±9F), and 200 rounds shall be tested at 70C ±5C (158F ±9F). Record any damage to the packaging and cartridges. After completion of this test, 200 each of the same cartridges shall be subjected to the 1.5 meter, loose cargo, and 2.1 meter drop tests IAW TOP 4-2-602. After completion of these tests the cartridges shall be fired in 10 round bursts in an M2 BMG with M3 RA. The cyclic rate shall be recorded and any cartridge failures shall be recorded.

4.6.29.2 12.2 meter drop test. Four hundred cartridges packed IAW paragraph 5.2.1 shall be subjected to the 12.2 meter drop test IAW TOP 4-2-601. Two hundred rounds shall be tested at -50C ±5C (-58F ±9F), and 200 rounds shall be tested at 70C ±5C (158F ±9F). Record any damage to the packaging and cartridges.

4.6.30 Toxic fumes. The cartridges shall be fired in 10 round bursts in an M2 BMG with M3 RA. The muzzle of the weapon shall protrude from a 2130mm x 2130mm x 2130mm chamber. The air within the chamber shall be continuously monitored for carbon monoxide, ammonia, nitrogen dioxide, and sulfur dioxide. Three trials of 100 rounds each shall be performed. The test shall be repeated with Caliber .50 M33 Ball cartridges using the M2 BMG with the M3 RA removed.

4.6.31 Safety hazard classification. The test shall be conducted by the contractor at a facility of his choosing at his expense. The Government shall reserve the right to observe all testing. The tests shall be conducted IAW TB 700-2 with the following differences only one outer packing unit shall be placed on the stand and the stand shall be made of metal instead of wood for the external stack test (para 5-3g).

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4.6.32 Explosive sensitivity. Upon design submission the U.S. Army Safety Office shall determine which materials utilized in the cartridge require sensitivity evaluation testing by the contractor. The sensitivity tests required shall be as follows:

Test	Test Document	Test Procedure/ Paragraph
a. Friction test - apparatus and comparison values	JSSPM	5.3
b. Impact test - apparatus and comparison values	JSSPM/ TB 700-2	4.2 5.2i
c. Electrostatic discharge test - apparatus and comparison values	JSSPM	4.2
d. Auto ignition temperature	JSSPM	5.9
e. Explosive temperature (5 seconds)	MIL-STD-650	506.1
f. Detonation test	TB 700-2	5.2f
g. Ignition and unconfined burning test	TB 700-2	5.2g
h. Thermal stability test	TB 700-2	5.2h
i. Card gap test	TB 700-2	5.2j

5. PACKAGING

5.1 Preservation and packing. See 5.2.1.

* 5.2 Packaging.

* 5.2.1 Packaging Level A. The linked ammunition shall be packed IAW drawing 12576457. The M2A1 metal container shall contain 100 cartridges linked with M9 links. Two loaded M2A1 metal containers shall be packed in one wooden wirebound box IAW 12576456. Shipment shall be made IAW MIL-STD-129 and satisfy CFR TITLE 49 Parts 100-199 as applicable.

* 5.3 Marking and labeling. Containers shall be marked IAW MIL-STD-129, 12576456, and 12576457 and CFR TITLE 49 Parts 100-199 as applicable.

* 5.4 Unitized loads. The linked cartridges packed IAW 5.2 shall be unitized IAW AMC drawing 19-48-4116/14.

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6. NOTES

(This section contains information of a general or explanatory nature that may be be helpful, but is not mandatory.)

6.1 Intended use. The cartridges covered by this specification are intended for use in Caliber .50, M2 Browning Machine Guns with the M3 Recoil Amplifier, weapon system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Certificate of conformance for each lot or shipment of cartridges.

g. Type and level of packing for the cartridges.

h. Provisions for the submission of acceptance inspection reports containing the final inspection results for each lot of ammunition presented to the government (see 6.4).

6.3 Submission of contractor inspection equipment designs for approval. (See MIL-A-48078). Submit copies of designs as required to: Commander, ARDEC, ATTN: AMSTA-AR-QAC-C (D), Picatinny Arsenal, N.J. 07806-5000. This address will be specified on the Contract Data Requirements List, DD form 1423 in the contract.

6.4 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the US AMCCOM, one copy of all ballistic data IAW the Data Item listed on the Contract Data Requirements List, DD form 1423 in the contract, and the Ammunition Data Card for each lot shall be forwarded to: Commander, ARDEC, ATTN: AMSTA-AR-QAC-C (D), Picatinny Arsenal, N.J. 07806-5000.

6.4.1 Minimum propellant charge. The contractor is required to submit an analysis that establishes the minimum propellant charge required to preclude the possibility of a bullet in bore.

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6.5 Hazard notice. The cartridge described herein and some of the cartridge components are flammable and explosive and, therefore, present hazards in manufacture, handling, storage, and shipment. the contractor shall recognize these hazards and shall take appropriate measures to prevent fire, explosion, adverse environmental, rough handling, corrosive atmosphere, and electrically induced incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.6 Drawings. Drawings listed in Section 2 of this document under the heading US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, U.S. Army Armament Research and Development Command (ARRADCOM), or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under ARDEC.

6.7 Definitions.

6.7.1 Reliability. The number of successes (rounds chambered, fired, and ejected) divided by the number tested. Occasionally the last round of a continuous belt, due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure shall not be classified as a weapon stoppage nor shall this type of failure be recorded as a defect chargeable to the ammunition.

6.8 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, any proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.9 Subject term (key word) listing.

M2 Browning Machine Gun

M3 Recoil Amplifier

Machine Gun

Munitions

Small Arms Ammunition

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* 6.10 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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